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Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort is difficult to implement
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Quick Sort
- b. Insertion Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 0.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Merging
- d. Searching

The correct answer is: Sorting

Question **5**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **6**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Dividing the list into halves
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Using the divide-and-conquer approach
- b. Bubbling up the largest element to its correct position with each pass
- c. Being the most efficient sorting algorithm
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Greedy method
- c. Dynamic programming
- d. Backtracking

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It requires external libraries
- b. It only works with integer arrays
- c. It is less efficient than custom sorting algorithms
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Quick Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It slows down the process
- c. It makes the data unsorted
- d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data in a single pass
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It allows for quicker identification of duplicates
- c. It has no effect on the analysis process
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates